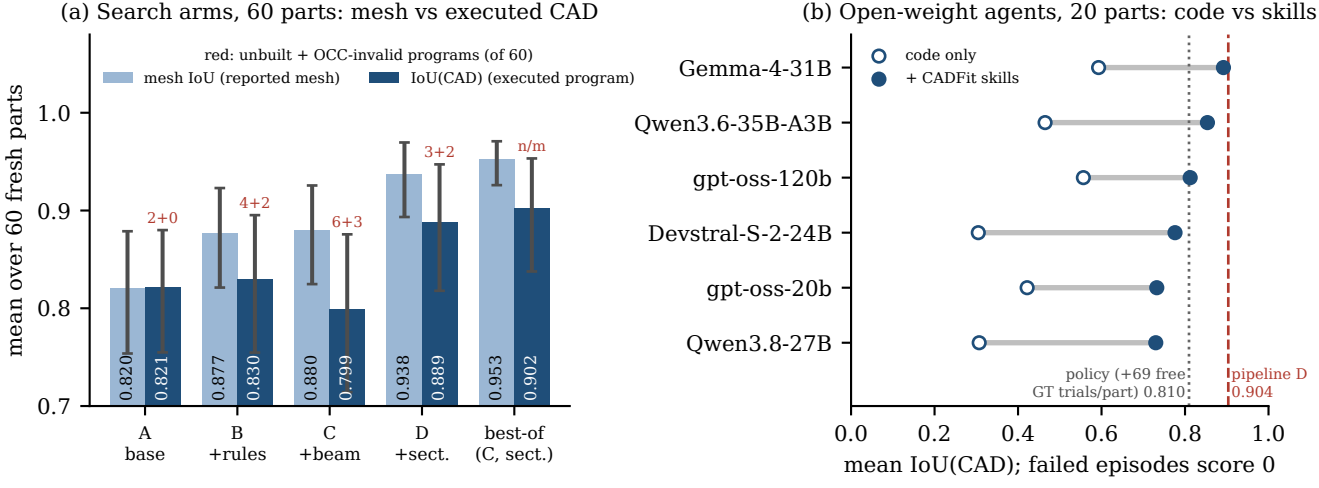
(a) Search arms, 60 parts: mesh vs executed CAD
red: unbuilt + OCC-invalid programs (of 60)
mesh IoU (reported mesh)
IoU(CAD) (executed program)
mean over 60 fresh parts
1.0
0.9
0.8
0.7
2+0
4+2
6+3
3+2
n/m
0.820
0.821
0.877
0.830
0.880
0.799
0.938
0.889
0.953
0.902
A
base
B
+rules
C
+beam
D
+sect.
best-of
(C, sect.)
(b) Open-weight agents, 20 parts: code vs skills
code only
+ CADFit skills
Gemma-4-31B
Qwen3.6-35B-A3B
gpt-oss-120b
Devstral-S-2-24B
gpt-oss-20b
Qwen3.8-27B
policy (+69 free GT trials/part) 0.810
pipeline D 0.904
0.0
0.2
0.4
0.6
0.8
1.0
mean IoU(CAD); failed episodes score 0